

7. [Maximum mark: 7]

Consider the function $f(x) = \sec\left(x - \frac{\pi}{4}\right)$, for $0 \leq x \leq \frac{\pi}{2}$.

(a) Determine the range of f . [3]

The region bounded by the graph of $y = f(x)$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{2}$ is rotated 2π radians about the x -axis.

(b) Find the volume of revolution generated. [4]

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